A Market-Adaptive Agent with Real-Time Scheduling for Maritime Spot Markets

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ABSTRACT

We propose an adaptive agent for the Maritime Shipping Competition. The agent must bid for transportation requests while scheduling vessels in real time. A key challenge is that request costs depend on the current fleet schedule. At the same time, bidding decisions must be made under uncertain and changing competition. We estimate scheduling-related costs using repeated schedule construction with different insertion orders for both pre-auction bidding decisions and post-auction fleet integration, providing more stable cost and feasibility estimates. We evaluate market conditions online from auction outcomes. We use moving statistics to track price-to-cost ratios. We also apply a probabilistic model to handle partially observed feedback, which is to adapt to unknown and varying market strength and adjust bidding behavior accordingly.

KEYWORDS

autonomous agents, maritime shipping, sealed-bid second price auctions, cost estimation, multi-start scheduling, online market learning, adaptive bidding

1 INTRODUCTION

In the maritime spot market, transportation contracts are usually allocated through sealed-bid auctions. In this setting, agents must bid for transportation requests while simultaneously scheduling their vessels under uncertainty.[10] Transportation requests arrive continuously and require fast price quotations together with feasible route planning. At the same time, vessel locations, capacity constraints, and time-window restrictions make the scheduling problem inherently complex. Under the strict time limits of the competition, and with much information remaining unknown, computing an optimal schedule is often impractical. The scheduling problem is computationally NP-hard, and in practice agents typically have to rely on approximate or heuristic methods to estimate execution costs and construct feasible deployment plans.

From a game-theoretic perspective, this problem can be viewed as a Bayesian game with incomplete information[8]. Each agent can only observe its own fleet state and cost estimates, while the cost structures and bidding behaviors of competitors remain unknown. In addition, the independent private value assumption that supports truthful bidding in reverse second-price auctions does not hold in this setting[2, 10]. Transportation costs across different requests are strongly interdependent, and winning only a subset of bids may even lead to overall losses. At the same time, the level of market competition is also unknown, and the scale at which other agents may place their bids is difficult to anticipate in advance.

Based on these challenges, this article sets two main goals. First, it does not try to predict the risk preferences or bidding strategies of competitors. Instead, it focuses on the overall level of market competition. The paper introduces a robust proxy for this purpose. This allows the agent to balance cautious and aggressive behavior and adjust to market changes. Second, we aim to develop a fast scheduling approach that can generate high-quality vessel deployment plans under strict real-time constraints, thereby reducing execution costs and providing reliable cost estimates to support bidding decisions.

The remainder of this report is organized as follows. Section 2 discusses the main challenges of the competition from a game-theoretic perspective and reviews related auction and scheduling methods. Section 3 then describes the overall decision framework of our agent and explains the key ideas behind its design.

2 THEORETIC CONSIDERATIONS FOR AGENT DESIGN

Algorithms closely related to maritime shipping competition primarily fall into two main sides: (i) Auction-Side; (ii) Scheduling-Side.

2.1 Auction-Side Considerations

2.1.1 Failure of Truthful Bidding. Since truthful bidding is no longer applicable, the *intrinsic cost* of an individual request cannot be precisely [7]. The dominant-strategy incentive compatibility of Vickrey auctions relies on the assumption of *single items* with *independent private costs*, which does not hold in transportation problems where costs are *state-dependent* and *interdependent* across requests[2, 8]. In such settings, individual cost revelation becomes unreliable and may expose the agent to *bundling risk*.

Our agent does not attempt to infer a true standalone cost for each request. Instead, it approximates request costs using *schedule-aware marginal insertion costs* obtained from multiple tentative schedules constructed via a *multi-start heuristic*[9]. These approximate costs serve as the basis for bidding decisions, allowing the agent to control combinatorial risk without relying on explicit information about competitors, which leads to more stable utility outcomes and reduces the likelihood of loss-inducing partial wins in repeated auctions.

2.1.2 Incomplete Information and Market Uncertainty. From a perspective of game theory,the environment of auction can be characterized as an *incomplete-information setting*. Each agent observes only its own *fleet state* and *internal cost estimates*, while competitors' *cost structures*, *bidding behavior*, and overall *competition intensity* remain unknown and may vary over time. Under such conditions, bidding strategies based on *fixed markups* or *static equilibrium assumptions* become unreliable[13].

To address this uncertainty, we adopt a lightweight adaptive bidding mechanism rather than a fixed pricing rule. Market price strength is approximated using an *exponentially weighted moving average (EWMA)*, bidding aggressiveness is regulated through a *feedback-based bid scaling factor*, and different pricing rules are applied depending on the approximated *market regime*. Also, we employ a *left-censored learning* approach to update market price beliefs when only the observable data is the *upper bound* of competitors' bids. Together, these mechanisms enable *robust market learning* and *adaptive bidding* under incomplete and noisy data while remaining computationally simple[6].

Regarding market modeling under uncertainty, several alternative approaches can be considered, which we briefly discuss below:

Offline Calibration. One approach is to assume a fixed bidding distribution that is calibrated offline from historical data[11]. While simple, this method relies on stationarity assumptions and cannot adapt to changing market conditions[12].

Explicit Opponent Modeling. Another alternative is to explicitly model competitors using parametric assumptions. Such models can be expressive but are sensitive to model mismatch and non-stationarity[5].

Left-Censored Learning. In this work, we have taken a left-censored learning approach that updates market price beliefs from partial auction feedback, where an upper bound on competitors' bids is observed[4]. This enables lightweight and robust learning under incomplete information[1].

2.2 Scheduling-Side Considerations

2.2.1 Bounded Rationality under NP-hard Scheduling. The underlying scheduling problem faced by the agent is very complex and is NP-hard. It must be solved under strict real-time limits. Under these conditions, the set of possible strategies is limited by high complexity and short decision time. Because of this, creating an optimal plan is usually not realistic. In this kind of environment, it is difficult to use precise optimization methods. This includes complex heuristic algorithms, accurate prediction, and similar approaches.

In response, we adopt a bounded-rational scheduling design rather than pursuing optimality. Specifically, the algorithm uses a multi-start insertion strategy. The agent can quickly build several workable plans and keep the *Top- K* of them. However, the agent does not rely on a single optimization method. Instead, it builds plans many times and extracts stable cost information from these runs. Based on this, the agent obtains a simple and reliable cost estimate. This approach works better in strategic situations with uncertainty.

2.2.2 Online and Post-Auction Scheduling Decisions. In this situation, each awarded contract changes the vessel's future cargo capacity and time schedule, based on the agent's actual costs. Because of this, scheduling decisions must be closely connected to the auction results. As a result, decisions made in one auction directly affect the overall planning of future auctions. If the agent treats bidding and scheduling as two separate tasks, this is a short-sighted approach. Similarly, focusing only on reducing transport costs while ignoring payments and penalties is also not appropriate. We therefore adopt a profit-driven post-auction scheduling strategy that is explicitly aligned with bidding decisions [3].

We therefore adopt a profit-driven post-auction scheduling strategy that is explicitly aligned with bidding decisions. The agent applies the same framework used in the bidding stage to the scheduling timetable. The schedule is then updated based on the received payments. The agent also considers two factors: additional execution cost and feasibility risk. This design keeps the bidding process consistent with the planning stage. At the same time, it links auction incentives with real operating limits. As a result, fewer awarded contracts turn out to be infeasible or unprofitable.

3 AGENT DECISION FRAMEWORK DESCRIPTION

In each auction round, our agent follows a fixed decision process that consists of four main steps. These steps define how the agent: (i) estimates costs, (ii) places bids, (iii) learns from the market, (iv) updates its fleet schedule.

The agent first estimates costs for incoming requests based on the current fleet schedule. It then decides which requests to bid on and at what prices. After the auction, the agent updates its view of the market and adjusts the fleet schedule to incorporate awarded requests. Details of these four steps are presented in the following subsections.

Figure **??** in Appendix B illustrates the detailed decision workflow of the agent within a single auction round.

3.1 Cost Approximation

Before bidding, execution costs are approximated using schedule-aware signals rather than optimal scheduling (Appendix A). A multi-start insertion heuristic generates multiple feasible schedules, from which Top-K solutions are selected to estimate marginal costs and stability signals (Algorithms 1 and 2).

3.2 Bidding Strategy

Bids are constructed from the schedule-aware cost signals obtained in Section 3.1, namely the marginal insertion cost $m(r)$ and stability score $p(r)$. Only requests with sufficiently high stability are considered for bidding.

For each candidate request, a blended cost estimate $\hat{c}(r)$ is formed by combining a direct cost proxy $C(r)$ with $m(r)$ (Appendix A). Market conditions are summarized by an EWMA-based price signal, which regulates bidding aggressiveness via a global scaling factor. Pricing follows a regime-dependent rule that adapts bids to tight, medium, and loose market conditions. The resulting bidding procedure is summarized in Algorithm 3.

Algorithm 1 GenerateCandidatePlans (Multi-Start Insertion)

Require: Request set R, initial fleet schedules S, number of starts M
Ensure: Candidate plans $\mathcal{P}$, per-request insertion cost lists $\{C(r)\}_{r\in R}$

```
Initialize candidate plan set P ← ∅
Initialize per-request cost list C(r) ← [ ] for all r ∈ R
for i ← 1 to M do
    Sample a random permutation π_i of requests in R
    S_i ← S;   scheduled_i ← ∅
    for all r ∈ π_i do
        (Δc, S_i) ← BestInsertion(r, S_i)
        if Δc ≠ ∞ then                    ▷ feasible insertion exists
            Append Δc to C(r)
            scheduled_i ← scheduled_i ∪ {r}
        end if
    end for
    Add (S_i, scheduled_i) to P
end for
return P and {C(r)}_{r∈R}
```

Algorithm 2 AggregateCostSignals (Top-K Marginal Cost & Stability)

Require: Candidate plans $\mathcal{P}$, cost lists $\{C(r)\}_{r\in R}$, Top-K size K
Ensure: Marginal cost estimates $\{m(r)\}_{r\in R}$, stability scores $\{p(r)\}_{r\in R}$

```
Select Top-K plans P_K ⊆ P (by feasibility and objective score)
for all r ∈ R do
    cnt(r) ← |{(S_j, scheduled_j) ∈ P_K : r ∈ scheduled_j}|
    p(r) ← cnt(r)/K
    if cnt(r) > 0 then
        m(r) ← Average(C(r))
    else
        m(r) ← ∞
    end if
end for
return {m(r)}_{r∈R}, {p(r)}_{r∈R}
```

Algorithm 3 Regime-Dependent Bidding

Require: Requests R, marginal costs $\{m(r)\}$, stability $\{p(r)\}$, market signal $\bar{R}$, bid scale α
Ensure: Bid set $\mathcal{B}$

```
s ← MarketStrength(R̄)
Θ ← Params(s)                    ▷ selection, weights, markups
C ← SelectRequests(R, m, p, Θ)
B ← ∅
for all r ∈ C do
    C ← DirectCostProxy(r)
    ĉ ← BlendCost(C, m(r), Θ)
    (b, f) ← PriceByRegime(ĉ, m(r), C, s, Θ, R̄)
    b ← max(αb, f)
    B ← B ∪ {Bid(r, b)}
end for
return B
```

3.3 Market Learning

Market conditions are learned online from repeated auction outcomes. The agent maintains a belief over the payment-to-cost ratio using an EWMA-based statistic and a discretized ratio distribution updated from partial auction feedback (Appendix 5, 9).

Since only upper bounds on competitors' bids are observable in second-price auctions, belief updates operate under left-censored observations. The learned market signal is used to estimate market strength and to adapt both bidding aggressiveness and regime selection over time (Appendix 8).

Algorithm 4 Post-Auction Market Update

Require: Ledger $\mathcal{L}$, last bids b_{last}, cost proxy $C(\cdot)$, ratio belief $\mathcal{F}$, market EWMAs $(\bar{r}_{mkt}, \bar{r}_{low})$, bid scale α
Ensure: Updated $(\mathcal{F}, \bar{r}_{mkt}, \bar{r}_{low}, \alpha)$

```
(R^ex, U^cen, R_mkt, R_low, w, b) ← ExtractRatios(L, b_last, C)
F ← UpdateRatioBelief(F, R^ex, U^cen)        ▷ exact + left-censored
(r̄_mkt, r̄_low) ← UpdateEWMAs(r̄_mkt, r̄_low, R_mkt, R_low)
ŵ ← w/max(1, b)                              ▷ win rate
α ← UpdateBidScale(α, ŵ, r̄_mkt)
return (F, r̄_mkt, r̄_low, α)
```

3.4 Post-Auction Scheduling

Post-auction scheduling reuses the same insertion-based feasibility checks as in Section 3.1, but applies them only to awarded requests. Moreover, insertion decisions are guided by the realized profit signal $U(r) = P(r) - \Delta C(r)$ (Appendix 3). Assignments that are infeasible or yield negative profit are discarded to avoid loss-inducing commitments.

Algorithm 5 Post-Auction Scheduling (receive())

Require: Awarded requests W, schedules S, payments $P(\cdot)$

```
for all r ∈ W do
    (ΔC, S') ← BestInsertion(r, S)
    if ΔC ≠ ∞ and P(r) − ΔC ≥ 0 then
        S ← S'
    end if
end for
return S
```

4 EXPERIMENTAL EVALUATION AND DISCUSSION

4.1 Performance Evaluation

We evaluate the robustness of the proposed agent under varying market conditions and fleet sizes. The agent competes against three opponents: two fixed-margin agents (*MyArchEnemy* and *TheScheduler*) and one adaptive-profit agent(*groupn1*).

Market competitiveness is controlled by varying the profit factors of the fixed-margin opponents from $(1.5, 1.5)$ to $(5, 5)$ and $(50, 50)$, while fleet sizes are varied from $(1, 1, 1)$ to $(3, 3, 3)$. For each configuration, we conduct 10 repeated runs and report the

Table 1: Performance summary (economic metrics, mean ± std over 10 runs)

Market	Fleet	Wins	Income	Penalty
(1.5, 1.5)	(1, 1, 1)	6.2 ± 1.3	−7098 ± 142	0
(5, 5)	(1, 1, 1)	20.3 ± 2.89	34351 ± 8430.66	0
(50, 50)	(1, 1, 1)	36.4 ± 1.26	494304 ± 34937	6807 ± 3316
(1.5, 1.5)	(3, 3, 3)	5.2 ± 0.8	4.8 ± 0.6	0
(5, 5)	(3, 3, 3)	19.33 ± 1.58	−3363 ± 243	0
(50, 50)	(3, 3, 3)	42.6 ± 1.5	789608 ± 1226	3178 ± 452

Market	Fleet	Revenue	Cost	Income/Win
(1.5, 1.5)	(1, 1, 1)	6518 ± 142	13616 ± 167	−1183 ± 23
(5, 5)	(1, 1, 1)	66947 ± 11239	32595 ± 288	1671 ± 197
(50, 50)	(1, 1, 1)	559730 ± 37623	58618 ± 416	13733 ± 948
(1.5, 1.5)	(3, 3, 3)	9434 ± 775	34619 ± 126	−5037 ± 156
(5, 5)	(3, 3, 3)	48762 ± 459	52125 ± 512	−173 ± 11
(50, 50)	(3, 3, 3)	893414 ± 4526	100627 ± 5218	18362 ± 688

mean and standard deviation of key metrics, including wins, income, revenue, cost, penalty.

Table 1 summarizes performance across market conditions and fleet sizes.

Table 1 shows that the proposed agent exhibits stable performance across different market conditions and fleet sizes. Both the number of wins and cumulative income vary only slightly across repeated runs, indicating low sensitivity to stochasticity in request arrivals and auction outcomes.

At the same time, performance differences across environments are clearly observable. In particular, the number of awarded contracts and total income change with market competitiveness and fleet size, suggesting that the agent adapts its bidding behavior to different operating regimes rather than following a fixed pricing policy. These results support the robustness and adaptivity of the proposed strategy.

No penalties are observed under low profit factors, suggesting that the scheduling strategy reliably preserves feasibility in tight markets. Penalties arise only in highly loose markets, reflecting an intentional risk–reward trade-off where increased revenue from additional wins outweighs penalty costs.

The income per win, remains consistently positive with low variance, suggesting that the agent reliably identifies profitable requests and maintains accurate cost estimation across environments.

A clearer indicator plot and the corresponding changes in variables are shown in Figure 2 and Figure 3 in Appendix B.

4.2 Design Insights and Debugging

Cost Signal Balancing. In early experiments, the agent relies too heavily on marginal insertion costs. When the market got tight, it became overly cautious and often did not place any bids.

We then put more emphasis on absolute cost signals in these cases. That pushes the agent to submit low-risk bids that should not lose money, so it avoids sitting idle while still staying conservative.

Stability via Multi-start Sampling. Using only a few tentative schedules caused a lot of variability, mostly because the insertion order was random.

Sampling a larger Top-K set makes the marginal cost estimates steadier, and the feasibility and profit signals come out more consistent—assuming there is enough compute time.

Market Regime Identification. Bidding behavior depends a lot on how we split the market into different regimes. By tuning the thresholds used to judge market strength, the agent can better spot situations where revenue is likely to beat execution cost. That leads to more sensible switches between conservative and aggressive bidding.

Simplified Cost Modeling. A cost model that was too complicated added extra noise when time was tight. By simplifying the execution cost proxy, bid prices became more stable and overall income increased. This suggests that in real-time decisions, simpler cost estimates can actually work better.

4.3 Limitations and Critical Discussion

The proposed agent uses a heuristic scheduling approach and a sampling-based cost estimation method. In other words, in situations where there are a large number of vessels or a significant increase in the number of requests, its performance may decline. The agent works well only when enough computing time is available.

Market learning is based on overall price signals rather than on explicit modeling of the behavior of competitors. While this design improves robustness under non-stationary conditions, it may react slowly to sudden shifts in competitor behavior or market dynamics.

The strategy adopts an explicit risk–reward trade-off in highly loose markets, where penalties are tolerated in exchange for increased revenue. This assumption may not hold if penalty mechanisms or enforcement policies change, and further adaptation would be required in such settings. Moreover, before learning about the signal, in the initial rounds, the agent is conservative by default, potentially missing out on high-quality opportunities in those early rounds.

An additional limitation was observed in extreme market conditions. A limitation arises where a single agent dominates most contracts. In such cases, the estimated market strength remains near zero, causing the agent to remain in a conservative regime even when external bids are high. Future work may explore alternative scaling schemes that remain informative under high market outcomes.[placeholder]: 2219 words in total

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A ADDITIONAL FORMULATIONS

Variables

Symbol	Description
r	transportation request
v	vessel
Π_v	current schedule of vessel v
$P(r)$	payment received for request r
$C(r,v)$	direct cost proxy of serving r by v
$\Delta C(r,v)$	marginal insertion cost of r into Π_v
$U(r,v)$	profit signal
$U'(r,v)$	risk-adjusted utility
$R(r,v)$	market price–cost ratio
$b(r)$	bid for request r
$p(r)$	stability score (Top-K appearance frequency)
α	global bid scaling factor

Cost and Profit Signals

$$C(r,v) = C_{\text{travel}}(r,v) + C_{\text{handling}}(r,v) + C_{\text{reposition}}(r,v) \tag{1}$$

$$\Delta C(r,v) = \max(\Delta C_{\text{sched}}(r,v),\ C(r,v)) \tag{2}$$

$$U(r,v) = P(r) - \Delta C(r,v) \tag{3}$$

$$U'(r,v) = U(r,v) - \lambda \max\left(0,\ \frac{\Delta C(r,v)}{C(r,v)} - 1\right) \tag{4}$$

Market and Bidding Signals

$$R(r,v) = \frac{P(r)}{C(r,v)} \tag{5}$$

$$\widehat{C}(r) = w \cdot C(r,v) + (1-w) \cdot \mathbb{E}[\Delta C(r,v)] \tag{6}$$

$$b(r) = \max\left(\widehat{C}(r)(1+\mu),\ \underline{b}(r)\right) \tag{7}$$

$$s = \text{clip}\left(\frac{\bar{R} - R_{\min}}{R_{\max} - R_{\min}},\ 0,\ 1\right) \tag{8}$$

$$\bar{R}_t = (1-\alpha)\bar{R}_{t-1} + \alpha\,\tilde{R}_t \tag{9}$$

$$p(r) = \frac{1}{K}\,|\{\pi \in \mathcal{P}_K : r \in \pi\}| \tag{10}$$

$$b(r) \leftarrow \alpha \cdot b(r), \qquad \alpha > 0 \tag{11}$$

B ADDITIONAL FIGURES

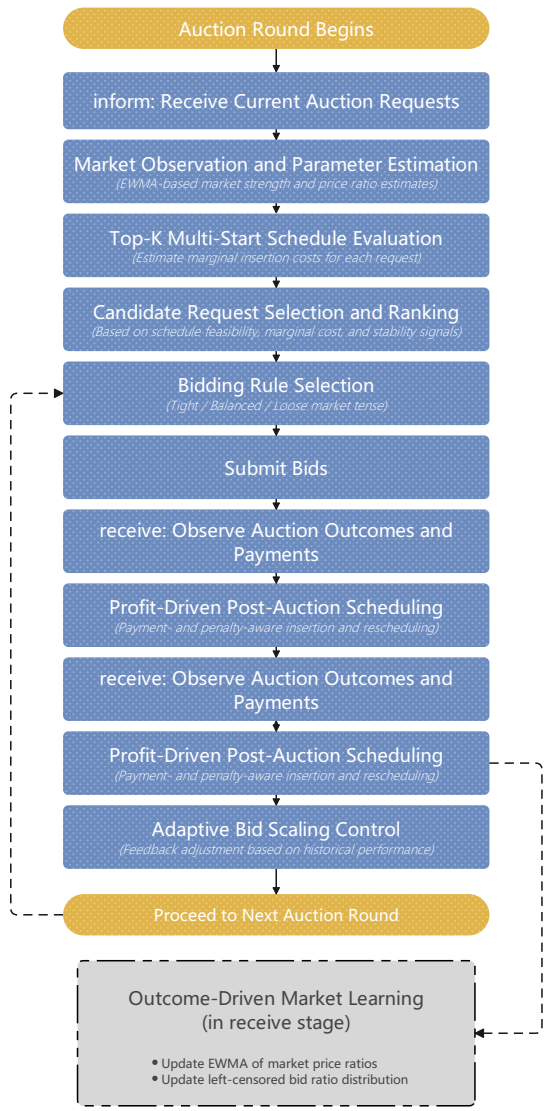

Figure 1: Overview of the agent's decision process within a single auction round.

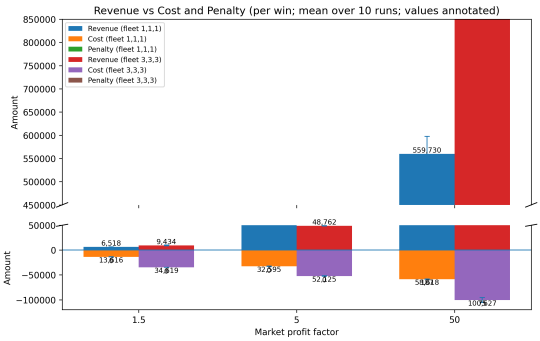

Figure 2: Revenue vs Cost and Penalty (per win; mean over 10 runs; values annotated).

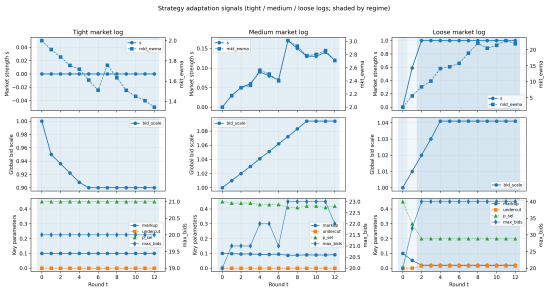

Figure 3: Strategy adaptation signals (tight / medium / loose logs; shaded by regime).